

1. Alignment ambiguity & Pipelines

Exercise #1: Setup, mafft, HoT

2. Definition of homology and *true* alignment

3. How to score how good an alignment is?

- Should have a plausible evolutionary history
- Additive scores are log probabilities
- Aligning “by eye”

4. Progressive Alignment

- Algorithm
- Guide Tree
- Score Function: Sum of Pairs

5. Refinement

Exercise #2: muscle

6. Probabilistic (pairwise) alignment

- most probability alignment, as opposed to...
- highest expected accuracy

Exercise #3: probcons and SuiteMSA

7. FSA

- Problems with Expected Accuracy (EA) ; Alignment Metric Accuracy (AMA)
- Annealing
- Estimate parameters

Exercise #4: FSA

8. PRANK & Tree Alignment

- Score: likelihood, tree based
- Progressive alignment: modified
- No profile alignment: reconstructs ancestors

Exercise #5: Prank

9. BALi-Phy & Joint Estimation (Tree & Alignment)

- Score: likelihood, tree-based
- Estimate parameters
- MCMC – no progressive alignment, automatic refinement
- Infers ancestors

BALi-Phy Tutorial